

Imp

Covariance & Correlation

:- Covariance is a Statistical term that refers to a systematic relationship between two random variable.

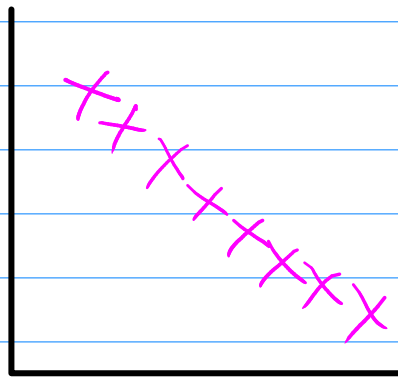
:- If one variable increase then the other variable increase or decrease.

:- It ranges between $-\infty$ to $+\infty$.

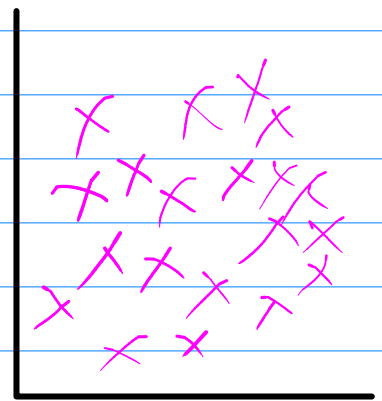
sq. feet.



as $y \uparrow$ $x \uparrow$ +ve Covariance.



$x \rightarrow y \downarrow$ -ve Covariance



Zero Covariance.

How much? $-\infty$ to $+\infty$

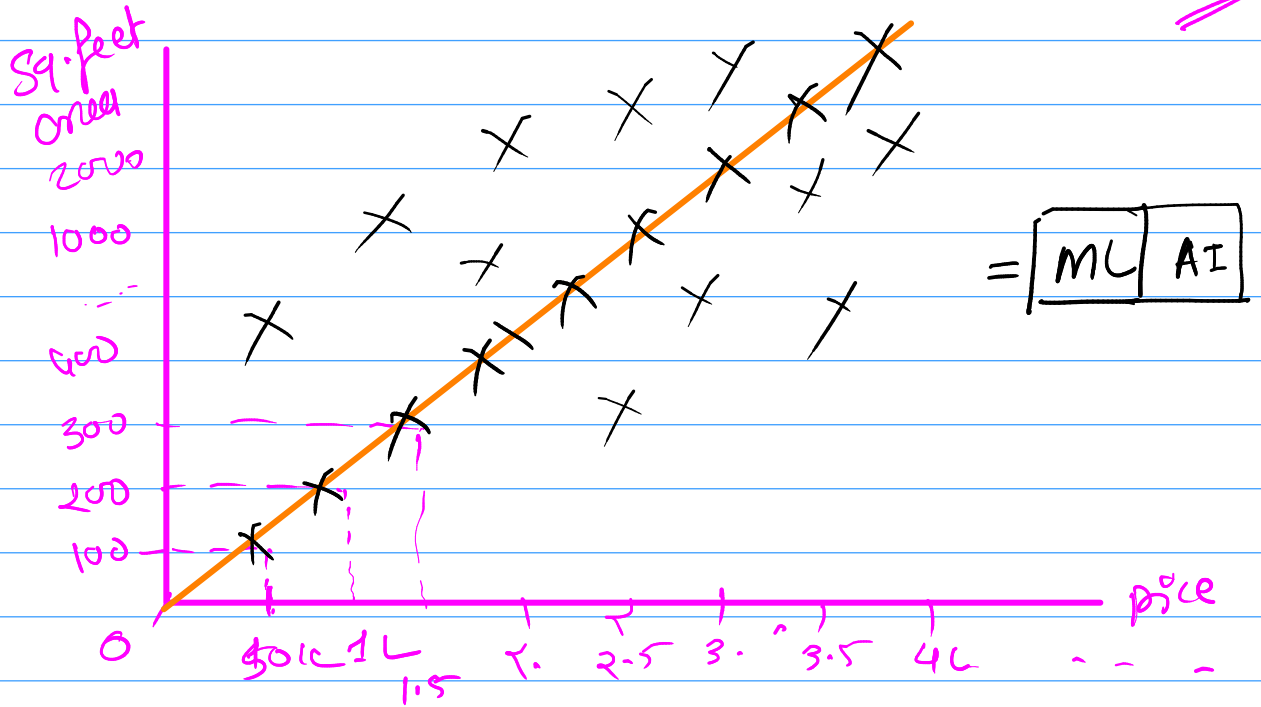
+30%, +300%,

Correlation Coefficient

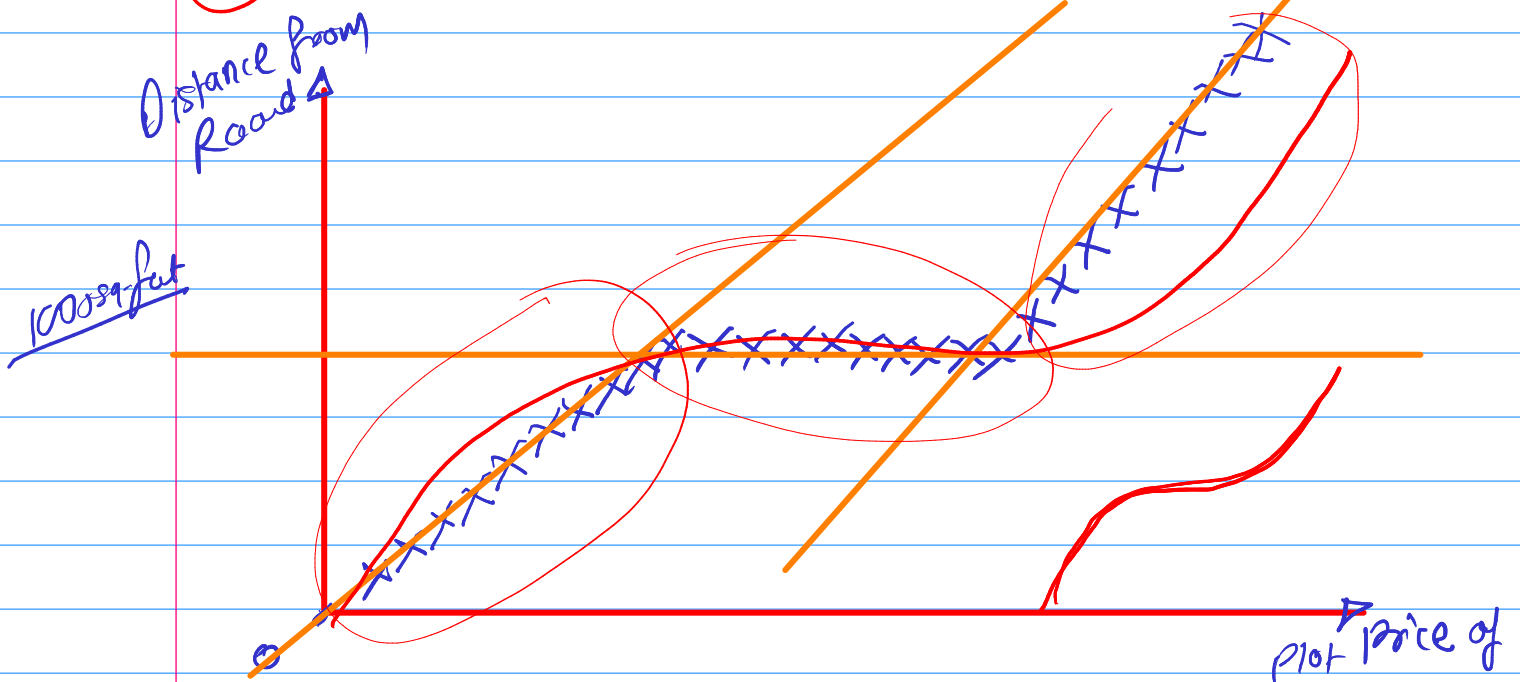
Range $\rightarrow 100\%$ to $+100\%$
 -1 to $+1$

① pearson Correlation ✓

$\rightarrow$ when we have to measure linear relationship between two variable. linear



② Spearman's Rank Correlation -1 to $+1$



$\rightarrow$ non linear Relationship then we use Spearman's Correlation.

Probability

It is the likelihood of possibility chances of the Random Variable

Variable ? $\Rightarrow$ Algebraic Variable

2 Random Variable

Whole No

Discrete Random Variable

(Continuous) Random Variable

2.5, 2.9, 3.7

Algebraic Variable

$$x + 5 = 10$$
$$x = 10 - 5$$

Variable

$$x = 5$$

It is a container that store the value.

2 Random Variable

It is a set of random value that stored in variable.

$$x = \{H, T\}$$

Coin toss

$$x = \{1, 2, 3, 4, 5, 6\}$$

Die throw



Continuous RV = $\{0 \text{ to } 10\}$ 0.1, 1, 2, 2.3.

Probability $\rightarrow$ It is likelihood or chance of event happening.

ex = Tossing a Coin

$X = \{H, T\} = 2 \text{ outcomes}$

$$P(H) = \frac{1}{2} = 0.5 = 50\%$$

$$P(T) = \frac{1}{2} = 0.5 = 50\%$$

$$\begin{aligned} \text{Total probability} &= P(H) + P(T) \\ &= 0.5 + 0.5 \\ &= 1 \end{aligned}$$

So Total probability is always 1

Probability Distribution

$\Rightarrow$ It is a list of all the possible outcome of a random variable along with their corresponding probability.

1	2	3	4
$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

probability distribution

Probability Distribution of single coin Toss.

event	H (1)	T (1)
probability	$\frac{1}{2}$	$\frac{1}{2}$

one Die

<u>Die</u>	1	2	3	4	5	6
<u>Prob</u>	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

1 -

What if two dice thrown at a time.

Dice 1 \ Dice 2	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$P(2) = 1/36$$

$$P(7) = 6/36$$

$$P(11) = 2/36$$

$$P(3) = 2/36$$

$$P(8) = 5/36$$

$$P(12) = 1/36$$

$$P(4) = 3/36$$

$$P(9) = 4/36$$

$$P(10) = 3/36$$

$$P(5) = 4/36$$

$$P(6) = 5/36$$

What if three dice thrown at a time?

In many scenarios, the number of outcome can be much larger and hence table would be tedious to write down (worse).

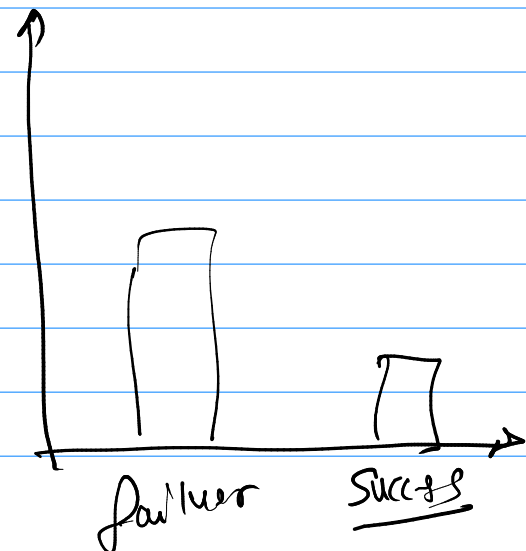
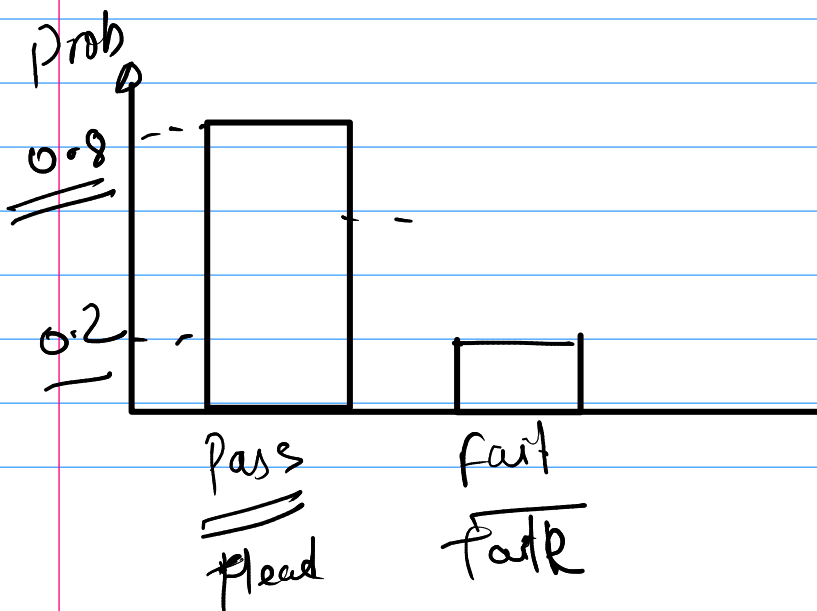
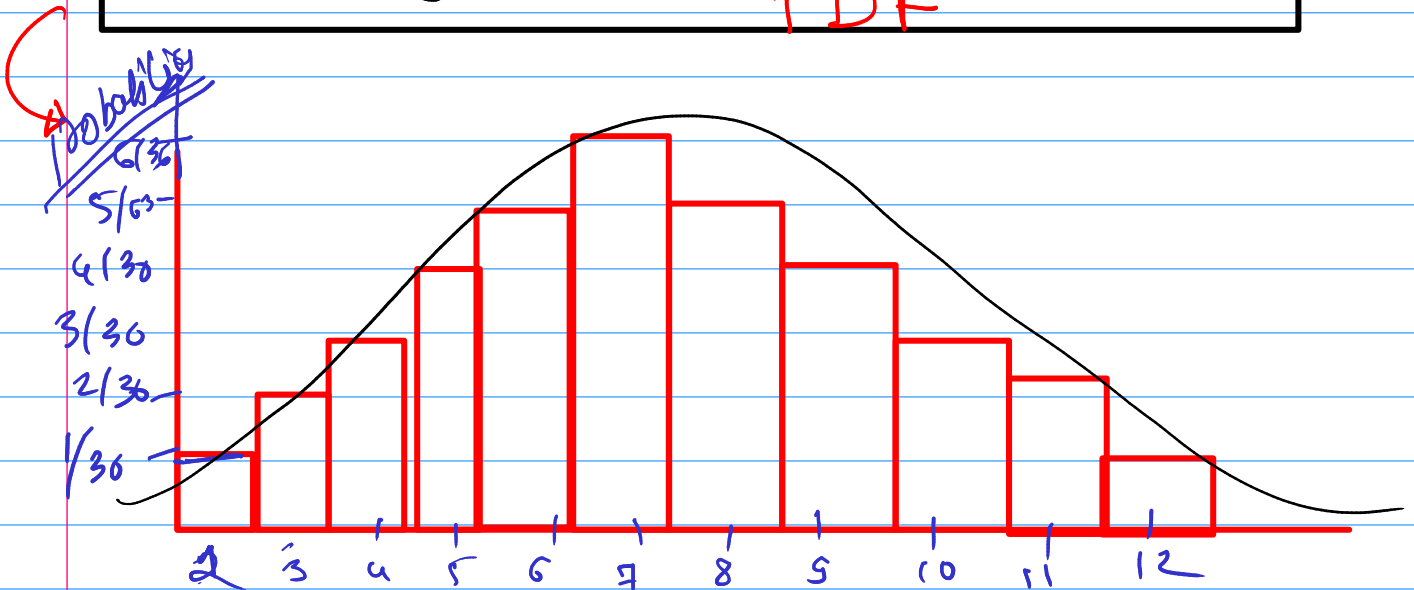
ex. height of people

What is $30/1^n$?

$\Rightarrow$ what if we use mathematical function to model the relationship between outcome and probability.

$$y = f(x) \begin{cases} x = \{1, 2, 3, 4, 5, 6\} = 1/6 \\ 0, \text{ otherwise} \end{cases}$$

Probability Distribution Function
PDF



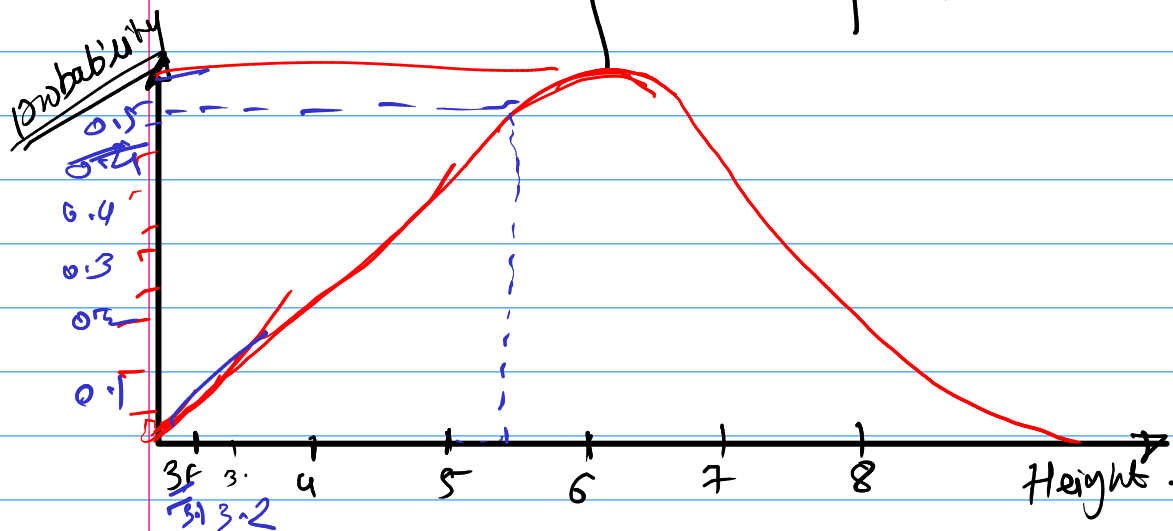
PDF $\Rightarrow$ probability Distribution Function

$\Rightarrow$ A PDF is a mathematical function that describe the probability of obtaining different value of a random variable in a Particular probability Distribution.

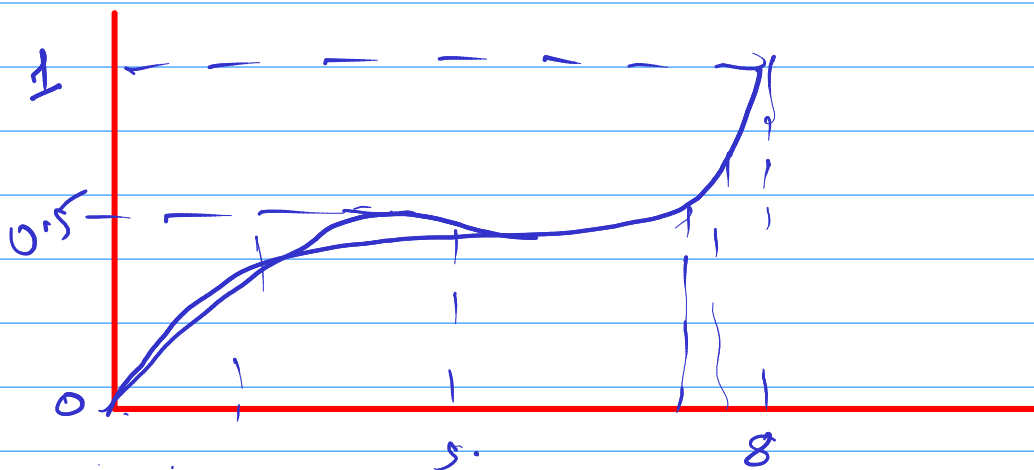
① Continuous Random Variable

PDF $\Rightarrow$ Probability Density Function

BP
hum.
Height
weight
temp



CDF $\Rightarrow$ Cumulative Density Function.

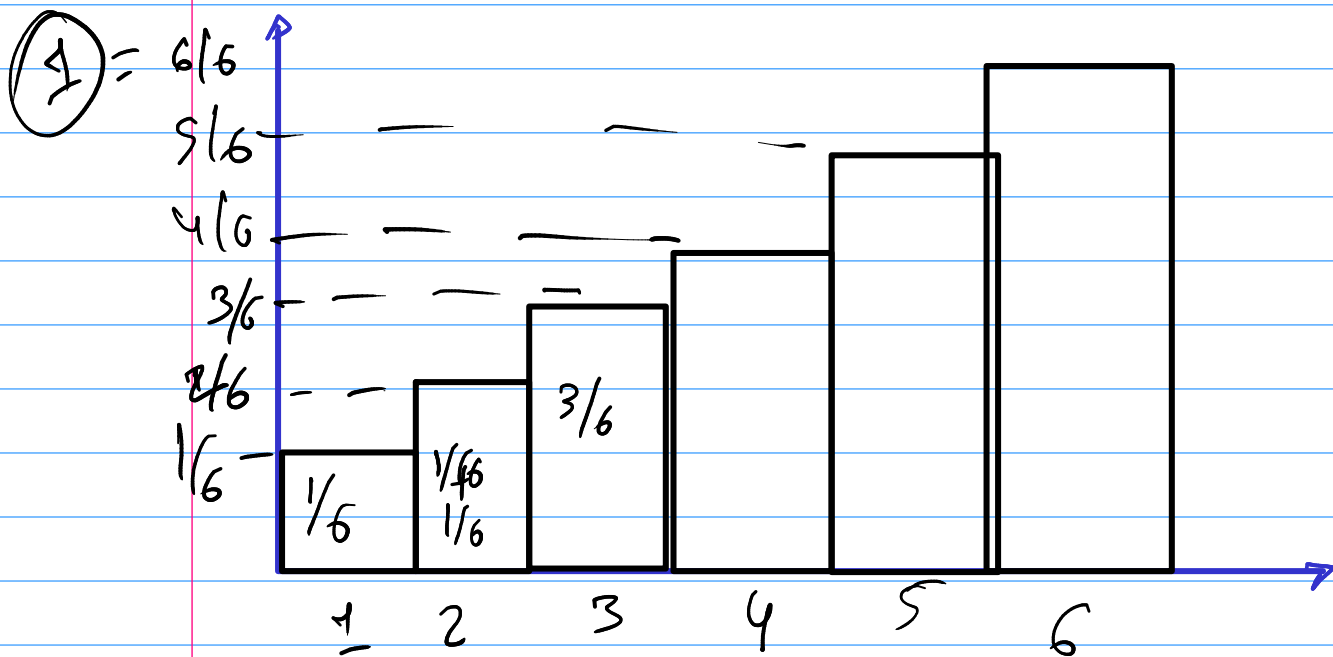


② Discrete Random Variable

PMF $\Rightarrow$ Probability Mass function



CMF $\Rightarrow$ Cumulative Mass function



Type of Distributions